

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50

Annexure A

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$**
- b. Choose $e=7$ and find the private key d**
- c. Encrypt the message $M=9$**
- d. Decrypt the ciphertext and verify the result**
- e. Analyze how changing e affects security**

Given

- $p=11$
- $q=13$
- $e=7$
- Message $M=9$

(a) Compute n and $\phi(n)$

Formula

$$n=p \times q$$

$$n=11 \times 13=143$$

Euler's Totient Function

$$\begin{aligned}\phi(n) &= (p-1)(q-1) \\ &= (11-1)(13-1) \\ &= 10 \times 12 = 120\end{aligned}$$

Answer

$$n = 143$$

$$\phi(n) = 120$$

(b) Find the Private Key ddd

Given

$$e = 7$$

Find d such that

$$ed \equiv 1 \pmod{120}$$

Try multiples

$$7 \times 103 = 721$$

$$721 \div 120 = 6 \text{ remainder } 1$$

Therefore

$$d = 103$$

Private Key

Public Key = (7,143)

Private Key = (103,143)

(c) Encrypt the Message M=9M=9M=9

Encryption Formula

$$\begin{aligned}C &= M^e \pmod{n} \\&= 9^7 \pmod{143}\end{aligned}$$

Calculation

$$\begin{aligned}9^2 &= 81 \\9^4 &= 81^2 = 6561 \\6561 \pmod{143} &= 126 \\9^7 &= 9^4 \times 9^2 \times 9 \\&= 126 \times 81 \times 9\end{aligned}$$

Modulo 143

$$C = 48$$

Ciphertext

Ciphertext = 48

(d) Decrypt the Ciphertext

Formula

$$\begin{aligned}M &= C^d \pmod{n} \\&= 48^{103} \pmod{143}\end{aligned}$$

Using modular exponentiation

$$M = 9$$

Decrypted Message

Original Message = 9

Verification

Encryption Successful

Decryption Successful

Recovered Message = 9

(e) Effect of Changing e

- A small value of e (such as 3) speeds up encryption but may reduce security if used improperly.
- Commonly used value **65537** provides a good balance between efficiency and security.

- e must satisfy:
 - $1 < e < \phi(n)$
 - $\gcd(e, \phi(n)) = 1$

Conclusion: Choosing an appropriate e improves security while maintaining efficient encryption.

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- a. Encrypt using ECB and CBC modes (Assume IV = 0000)
- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

Given

Plaintext Blocks

$P1 = 1010$

$P2 = 1010$

$P3 = 1111$

$P4 = 1010$

Key

$K = 1100$

Initialization Vector

$IV = 0000$

Assume encryption uses **XOR** with the key.

(a) ECB Encryption

Formula

Cipher = Plaintext XOR Key

Block 1

1010

1100

0110

Block 2

1010

1100

0110

Block 3

1111

1100

0011

Block 4

1010

1100

0110

ECB Ciphertext

0110

0110

0011

0110

CBC Encryption

Block 1

Plaintext XOR IV

1010

0000

1010

Encrypt

1010

1100

0110

Cipher1

0110

Block 2

1010

0110

1100

Encrypt

1100

1100

0000

Cipher2

0000

Block 4

1010

0011

1001

Encrypt

1001

1100

0101

Cipher4

0101

CBC Ciphertext

0110

0000

0011

0101

(b) Compare the Ciphertext Outputs

Bloc	EC	CB
k	B	C
P1	011 0	0110
P2	011 0	0000
P3	001 1	0011
P4	011 0	0101

(c) Which Mode Hides Patterns Better?

ECB Mode

- Each block is encrypted independently.

- Identical plaintext blocks produce identical ciphertext blocks.
- Patterns remain visible.
- Less secure.

CBC Mode

- Each plaintext block is XORed with the previous ciphertext block before encryption.
- Identical plaintext blocks produce different ciphertext blocks.
- Patterns are hidden.
- More secure.

Conclusion

CBC mode hides patterns better than ECB because each ciphertext block depends on both the current plaintext block and the previous ciphertext block, making repeated plaintext blocks produce different ciphertext outputs. This provides stronger protection against pattern analysis and improves overall security.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

		strong justification			
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation